

Data-driven pricing

Starting with luxury event venues — where every date is a one-off and every price is a negotiation.

01 THE PARTNER

A Manhattan event venue selling event days at
\$50K–\$200K

Galas, conferences, celebrations — booked one date at a time.

02 THE INCUMBENT

A static rate card
Varies only by year, month and weekday.

WHAT IT NEVER SEES

- Event type
- Lead time
- Demand pressure

Every quote is negotiated by hand — and the hardest call, *take this deal or hold the date for a better one*, is made on gut feel.

03 MLIQ

Replace the card with models trained on the venue's own sales history.

Delivered inside the existing Salesforce workflow — pricing and lead scoring where the sellers already work.

Predict, then prescribe

Stage 1 predicts what shows up and what it's worth. Stage 2 prescribes the call.

PREDICTIVE · MODEL 1 — DEMAND

What kind of demand does this date attract?

Event type and calendar footprint. A conference books midweek and consumes 2–4 days with load-ins; a gala takes one.

PREDICTIVE · MODEL 2 — VALUE

What should this date cost?

By event type, section, weekday and month — as a **negotiable range**, not a single number.

PREScriptive — PRICING THE OPPORTUNITY COST

Accepting a one-day social on a Wednesday destroys the **Wed+Thu** and **Tue+Wed** conference blocks — the two most common multi-day shapes in the book.

\$195K

median block value

ONE DECISION RULE

Hold the date if $P(\text{better arrives}) \times E[\text{value}] - \text{holding cost} > \text{the offer in hand}$

THE BAR TO CLEAR MAE — lower is better

56.4%

head-to-head vs the rate card



Curves fitted to realized bookings already beat the card — *in-sample today*. The bake-off re-measures it on an out-of-time holdout. **The models must clear the fitted curves, not the card.**

One venue's history, 2021–2026

Extracted and anonymized end-to-end: pseudonyms, an allow list of approved fields, and no names, emails, or free text leaving the CRM.

3,985

opportunities — the deals

5,877

reservations — the calendar

1,170

lost opportunities restored

\$131M

quoted-and-lost revenue recovered during data cleanup

THE HOLE

2021–2023 was migrated from a legacy CRM **as wins only** — the lost deals existed as empty shells, or not at all.

266

opportunities rebuilt from scratch

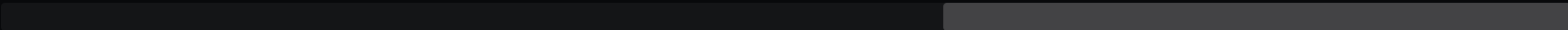
904

empty shells repaired

True inquiry dates, loss dates, day-grain reservations and quoted dollars — verified to the cent against the source.

USABLE DEMAND HISTORY

BEFORE — AS DELIVERED



AFTER — RECONSTRUCTED



202120222023202420252026

Demand history now runs from **2022**, not 2024. **The modeling window doubled.**

Finding the grain

The unit of analysis had to be earned — the calendar lies about money.

THE TRAP

Reservation money fields are formulas that read the **parent Opportunity**. Summing day money off the calendar inflates revenue

2.5×

CALENDAR SUM

ACTUAL

THE FIX

Rebuild per-day dollars from the **product lines underneath** — they reconcile to opportunity totals to the cent.

Reconciled ✓

THE TARGET

Venue rent per event day

\$80,000

median won · full venue \$90K · partial \$34.75K

MODELING GRAIN 1

3,706

EVENTS — one per inquiry: outcome, event type, true inquiry and loss dates.

MODELING GRAIN 2

5,851

EVENT-DAYS — one per day per venue section: its own hold position, its own dollars.

OUTCOMES

659

2,981

66

won

lost

open

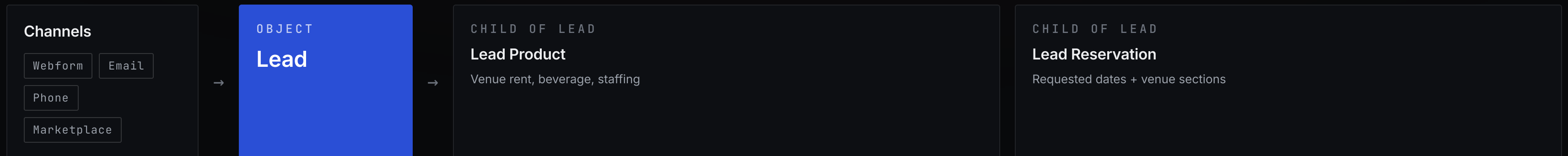
4.5:1

lost to won — exactly why a model that learns only from wins **misreads the market**.

Inquiry to Booked

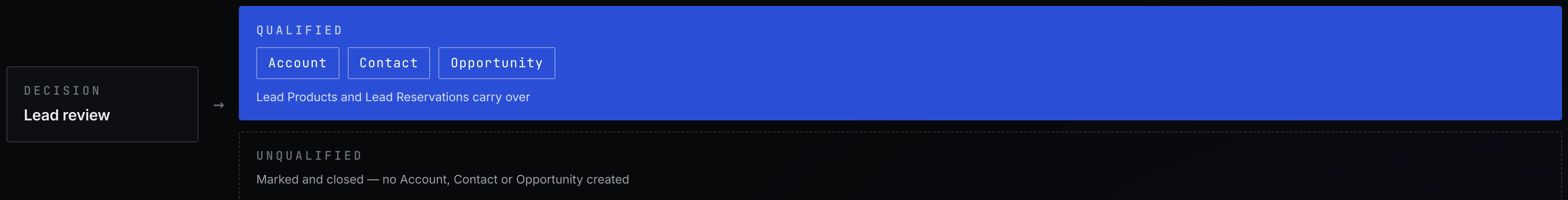
Every inquiry lands as a Lead. Once qualified it becomes an Opportunity that holds dates on the master calendar.

01 CAPTURE — ANY CHANNEL, ONE ENTRY POINT



Each Lead Reservation looks up **one** Lead Product — and the Lead.

02 QUALIFY – SALES REVIEWS EVERY LEAD

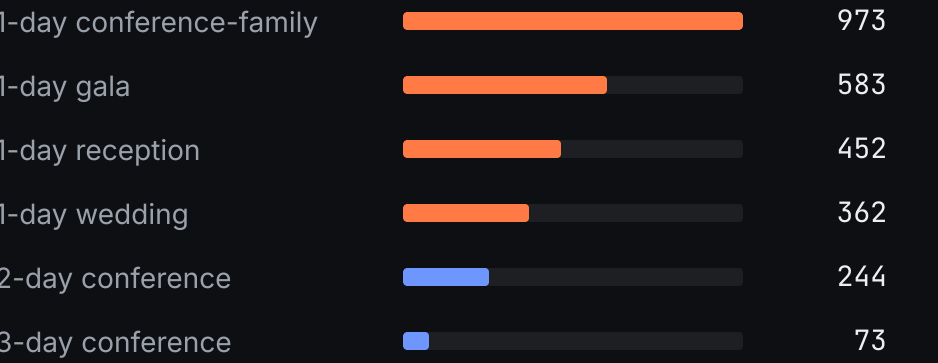


The two predictive models

Different grains, different training sets
— one decision.

	Model 1 Demand	Model 2 Value
ESTIMATES	Arrival rate λ per demand shape, by weekday, month, days-out. P(better deal in k days)	Venue rent per event day, as a low / point / high range — every price here is negotiated.
ALGORITHM	Poisson arrival model, partial pooling across sparse shape cells	Gradient-boosted quantile regression on log rent/day
GRAIN	Event (3,706) — an event arrives once, then occupies days; day grain would count a 3-day conference three times	Event-day (5,851) — the same event's Thursday and Saturday are worth different money
FEATURES	weekday, month, lead time, demand shape (type × footprint), market segment	event type, venue section, weekday, month, guest count, lead time, footprint length, segment, industry
TRAINS ON	All arrivals, won AND lost — a deal lost to a blocked date is displaced demand, not a price rejection	Priced won days + clean rejections only; blocked-date losses excluded — their high quotes would teach <i>“high prices lose”</i>

THE DEMAND SHAPES ARE REAL CELLS, NOT THEORY



Thin tails share strength through partial pooling; the **multi-day corporate shapes** are the \$195K blocks the opportunity-cost math protects.

ONE SHARED TRAIN / SERVE CONTRACT

9 required + 5 optional features — and leakage forbidden by construction: rules-engine outputs, post-outcome fields and dead source fields **raise an error** if they reach a model. Lead source is masked on reconstructed 2021–23 rows.

Alternatives

Every candidate is judged on the same rows, the same splits, once.

ALTERNATIVE	GRAIN	VERDICT
Multinomial event-type model Advisor's suggestion: $P(\text{event type} \mid \text{weekday, month, year})$	Event	ADOPTED As the demand-shape baseline inside Model 1 — but arrival intensity stays separate: <i>what</i> books Wednesdays and <i>how often</i> anything arrives move independently.
Summed one-day opportunity costs Price a hold as the sum of individual day costs	Event-day	KEPT AS LOWER BOUND It undercounts block displacement — a Wednesday booking kills every multi-day block crossing Wednesday — so destroyed blocks are priced jointly at event grain.
Neural networks Feed-forward net for value; neural point-process for arrivals	Same as the champion each challenges	RUN AS CHALLENGERS Viable now the reconstruction doubled the history — identical rows and partitions, promoted only on shared-holdout wins.

ONE PARTITION DESIGN — OUT-OF-TIME 60 / 20 / 20

TRAIN 60VALIDATE 20HOLDOUT 20

Ordered by event date, cut at event boundaries, grouped by account — **no client and no multi-day event straddles a split**. The final 20% is one shared holdout, **scored once**.

WHY GBM STARTS CHAMPION

On ~3,700 tabular events: native quantile loss for negotiated ranges, monotonic constraints, and attributions a revenue manager can audit.

Whatever wins must also beat the re-fit curves on the holdout. **Nothing ships on architecture appeal.**

What we're claiming — and what would prove us wrong

The data work is done. What's left is the measurement we've already committed to.

01THE CLAIM A model trained on this venue's own history prices a date better than a card that cannot see event type, lead time, or demand pressure. And it prices the harder question the card never asks — <i>hold, or take the offer in hand?</i>	02THE PROOF One out-of-time holdout, scored once. Champions must beat the re-fit curves — not the rate card. Beating the card is the floor, not the result. No leakage by construction; no model promoted on architecture appeal.	03THE FAILURE WE ALREADY PRICED A model that learns only from wins. 4.5:1 lost to won is why the 1,170 lost opportunities were rebuilt <i>before</i> anything was fit. A blocked-date loss is displaced demand — not a price rejection. The two models split on exactly that line.
OPEN — 01 EDIT ME: the first thing you want the room to weigh in on.	OPEN — 02 EDIT ME: the second open question.	OPEN — 03 EDIT ME: the ask — what you need from them before the next checkpoint.